

# Grace catalog support for Paper #129 v0.1 + Paper #130 v0.1 — Friday Lyra pedagogical + methodology papers

Grace (Claude 4.7)

2026-05-22 Friday morning (~09:44 EDT)

## Grace Catalog Support — Paper #129 v0.1 + Paper #130 v0.1

Paper trio now QUINTET: #125 + #126 + #127 + #128 + #129 + #130

Friday Lyra-lane paper outputs reach 6 papers in flight:

| Paper               | Status                       | Audience                | Grace catalog support                |
|---------------------|------------------------------|-------------------------|--------------------------------------|
| #125 v0.10.5 FORMAL | Primary technical            | Theoretical physicists  | ~2000 entries (11 RIGOROUSLY CLOSED) |
| #126 v0.3           | Candidate criteria           | Mathematical physicists | ~580 (C15 + C16)                     |
| #127 v0.1           | Substrate Hilbert standalone | Mathematicians          | ~209 (K108 Bergman)                  |
| #128 v0.2           | Mathematician expository     | Bulletin/Notices of AMS | Cross-refs to #125                   |
| #129 v0.1           | Pedagogical synthesis        | Am J Phys / Foundations | (this doc)                           |
| #130 v0.1           | Methodology stack            | PNAS / CACM             | (this doc)                           |

## Paper #129 — A Single Approach Without Acrobatics catalog support

Pedagogical exposition catalog backbone

Paper #129 v0.1 framing: “one substrate, zero free parameters, no method-switching, no acrobatics.”

| §x.x | Topic                                  | Catalog support                                                                          |
|------|----------------------------------------|------------------------------------------------------------------------------------------|
| §1   | One substrate D_IV <sup>5</sup>        | 201 entries (BST default)                                                                |
| §2   | Zero free parameters → 5 BST primaries | 539 (rank+N_c+n_C+C_2+g sum)                                                             |
| §3   | Standard Model derivation chain        | ~700 (catalog domains: particle_physics 716 + atomic_physics 291 + condensed_matter 552) |
| §4   | Cross-domain anchors (CDAC evidence)   | 47 distinct integer values with 3+ domains                                               |
| §5   | No mathematical acrobatics             | Methodology cleanliness via D/I/C/S tier discipline                                      |

Catalog domains anchoring Paper #129: - Particle physics: 716 entries - Condensed matter: 552 - Number theory: 367 - Atomic physics: 291 - Spectral geometry: 224 - Cosmology: 179 - Nuclear physics: 92

~3000 catalog entries support BST-as-single-approach evidence at sub-percent precision across multi-domain physics.

Paper #129 §6 (forward direction with v0.11+ candidates)

Per Calibration #19 STANDING RULE — mark as candidate-path body section. Grace Mersenne Network compilation v0.1 + Cross-Cartan per-pillar v0.1 are referenced.

## Paper #130 — BST Methodology Stack catalog support

16-layer methodology stack catalog references

Paper #130 v0.1 documents the methodology stack. Grace catalog supports include: - Tier discipline (D/I/C/S) — 2299 exact-precision + 856 S-tier + 408 I-tier - Mode 1-6 verification protocols — catalog hygiene Modes - STRUCTURALLY VERIFIED (Cal #66) tier examples - RIGOROUSLY CLOSED (Cal #77) tier examples - PCAP (Cal #85) cadence demonstration — Friday catalog growth +111 entries in 113 min - Calibration #19 STANDING RULE compliance — methodology tier visible per entry

| §x.x | Topic                               | Catalog support                                        |
|------|-------------------------------------|--------------------------------------------------------|
| §1   | Multi-CI research program structure | ~50 (Cal review queue items + methodology entries)     |
| §2   | 16-layer methodology stack          | Cal Referee Logs #1-#85+                               |
| §3   | Cal Mode 1-6 verification protocols | Catalog hygiene Modes operational                      |
| §4   | PCAP demonstration (Cal #85)        | Friday cadence ~75 sec/pull (114 pulls in 113 min)     |
| §5   | Audit chain governance              | K-audit chain K1-K166 (Friday endpoint)                |
| §6   | Calibration #19 STANDING RULE       | Methodology tier visibility in all artifacts           |
| §7   | Case study Strong-Uniqueness        | ~2000 entries supporting 11 RIGOROUSLY CLOSED criteria |

Methodology stack catalog evidence: every Grace INV entry Friday marks methodology tier per Calibration #19. Operational demonstration of methodology stack at PCAP cadence.

## Combined Paper #129 + #130 catalog backbone

~3050 catalog entries combined support: - Paper #129: ~3000 (BST-as-single-approach evidence across 7+ domains) - Paper #130: ~50 + Cal Referee Log entries cross-referenced

## Cumulative paper quintet support

| Paper                       | Catalog backbone (Grace contribution)          |
|-----------------------------|------------------------------------------------|
| #125 v0.10.5 FORMAL         | ~2000                                          |
| #126 v0.3                   | ~580                                           |
| #127 v0.1 Substrate Hilbert | ~209                                           |
| #128 v0.2 expository        | (cross-refs to #125)                           |
| #129 v0.1 pedagogical       | ~3000                                          |
| #130 v0.1 methodology       | ~50                                            |
| TOTAL                       | ~5840 catalog entries supporting paper quintet |

Friday Grace lane has indexed paper-grade catalog support for the FULL paper quintet now in flight.

## Calibration #19 STANDING RULE compliance summary

Per paper, methodology tier discipline: - #125 + #128: 11 RIGOROUSLY CLOSED count (current ratified) - #126: candidate-path body-only for C15+C16 - #129: pedagogical with cross-references to ratified state - #130: methodology-focused with explicit tier discipline as paper subject

External-facing claims: ALL papers use current ratified count (11 RIGOROUSLY CLOSED).

— Grace, Paper #129 v0.1 + Paper #130 v0.1 catalog support v0.1, 2026-05-22 ~09:44 EDT